Compositional system

geometry and dynamics for,

ESA Planck cosmic energy budget × redshift

z=0.0 → z=1100.0

Report title	Compositional system geometry and dynamics for, ESA Planck cosmic energy budget ×
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	esa_planck_cosmic
T (records)	17
D (carriers)	5
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	040d0e79dacf1ab6216a1a9141d59000…
Closed data SHA-256	40674b091a6338fac161aaac2d0ada6f…
Content SHA-256 (canonical JSON)	837bd6dec4c3d70addb670389f2864fc…

Reading order: geometry first, dynamics second.

```
§A  Data disclosure & processing notes
§B  Geometry : structure on the simplex
            evolution of proportions → ternaries → variation
            → correlation → scree → biplot → top-N biplot
            → balance-dendrogram → SBP table
§C  Dynamics : evolution through compositional space
            trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
            → bearing rose → bearing heatmap → ω → metric trace
            → helmsman → ring class → locks → CBS
            → pairwise divergence ranking → ω↔Hs
§D  Summary
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : esa_planck_cosmic_input.csv
  source SHA-256     : 040d0e79dacf1ab6216a1a9141d59000cdc6f2e800b842bfdb511304c5afaa62
  closed data SHA-256: 40674b091a6338fac161aaac2d0ada6f9a721eff140231c684dd8c26a37f00b5
  records (T)        : 17
  carriers (D)       : 5
  carriers           : Dark Energy, Cold Dark Matter, Baryons, Photons, Neutrinos

ORDERING
  is_temporal        : False
  ordering_method    : by-label
  caveat             : Order is treated as arbitrary; angular_velocity and energy_de
                       ly_depth are ordering-dependent.

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)
```

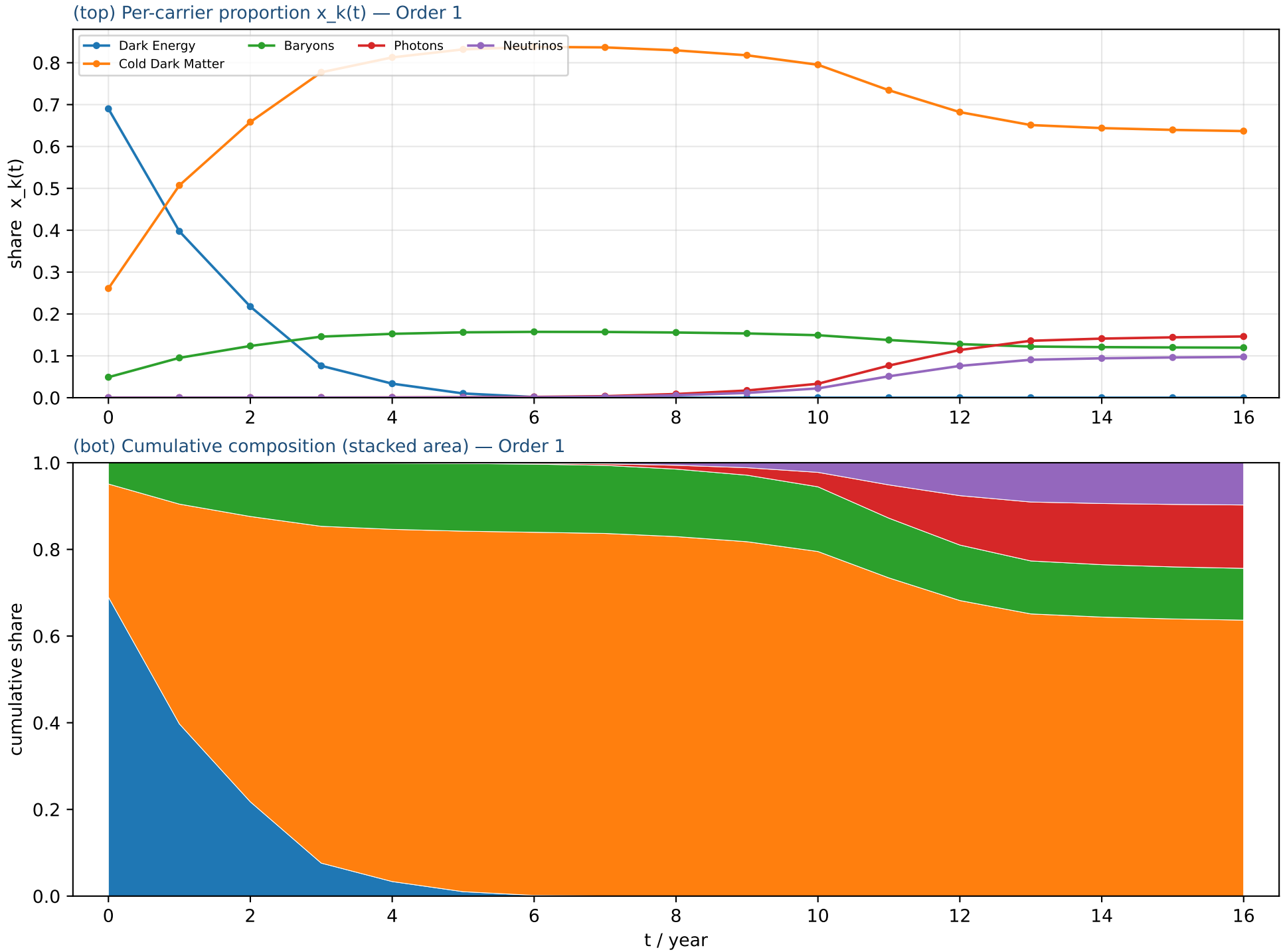
```
ENGINE SOURCE FINGERPRINT
  stage2_locked.py      : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  cnt.py                : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  cnt.cfg.sha256        : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  ⚠ {'flag': 'small_T', 'severity': 'warning', 'message': 'Trajectory too short fo
  ⚠ {'flag': 'pre_aligned_compositionally', 'severity': 'warning', 'message': 'Rec

LOCK EVENTS  (period-2 detector)
  total events       : 7
  lock_depth         : 7

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 0.7644726557235468
  M_sweep               : [{'M': 2, 'n_blocks': 9, 'H_mean_decimated': 0.782147023
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-4.44e-15, +3.55e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

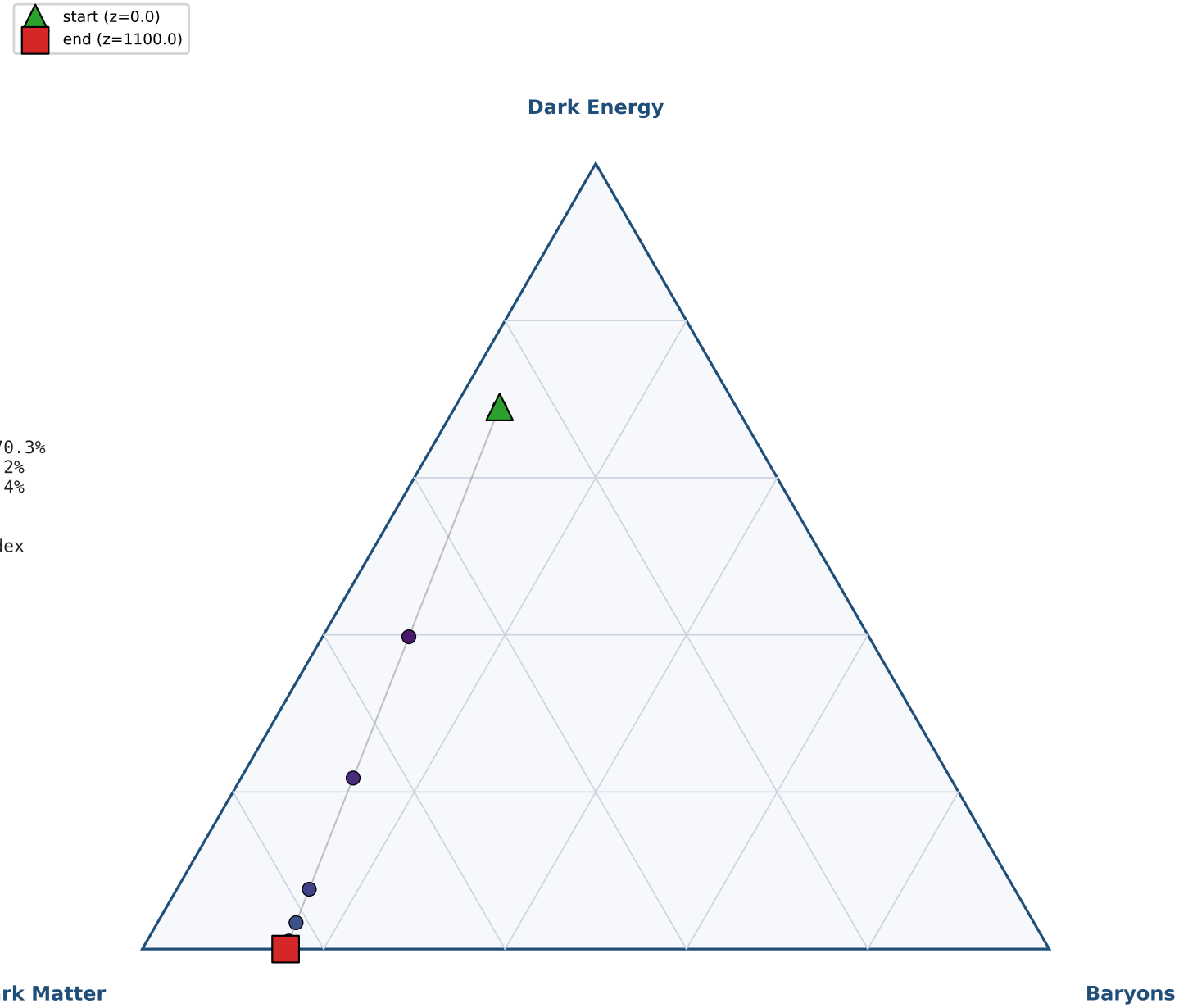


Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over Cold Dark Matter, Baryons, Dark Energy
(remaining 2 carriers absorbed into closure)

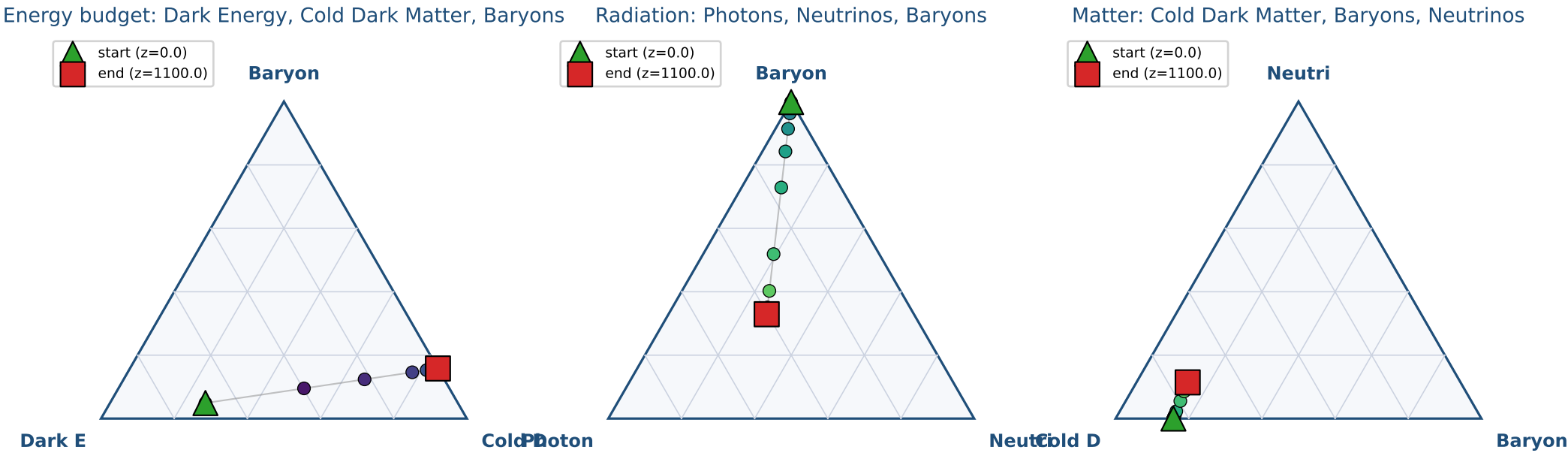



```
Top-3 by mean share:
  Cold Dark Matter  70.3%
  Baryons         13.2%
  Dark Energy      8.4%

T = 17 timesteps
colour map = time index
▲ start  ■ end
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



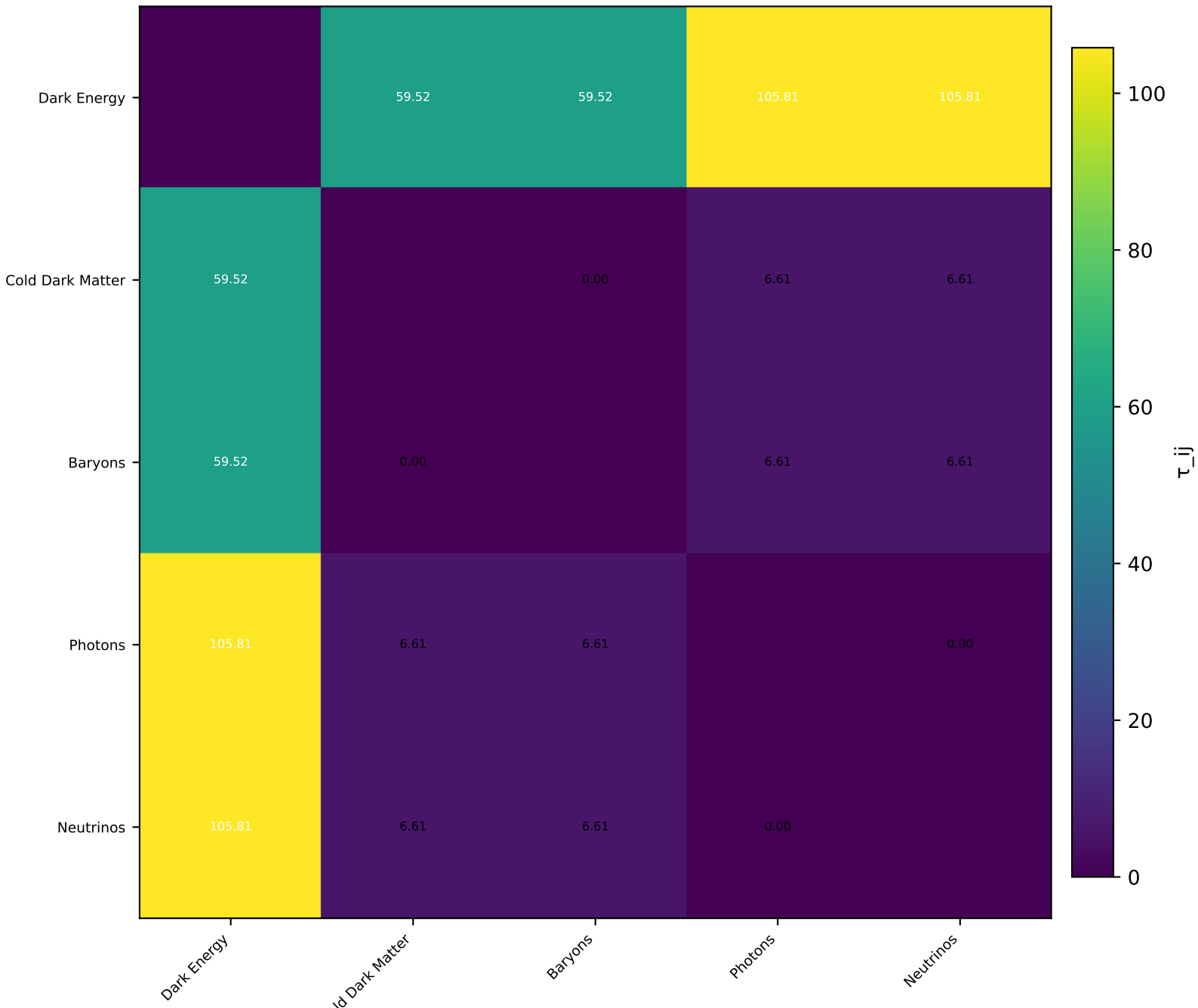
Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: user-configured (master_control.json)

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 6 | Variation matrix τ_ij heatmap

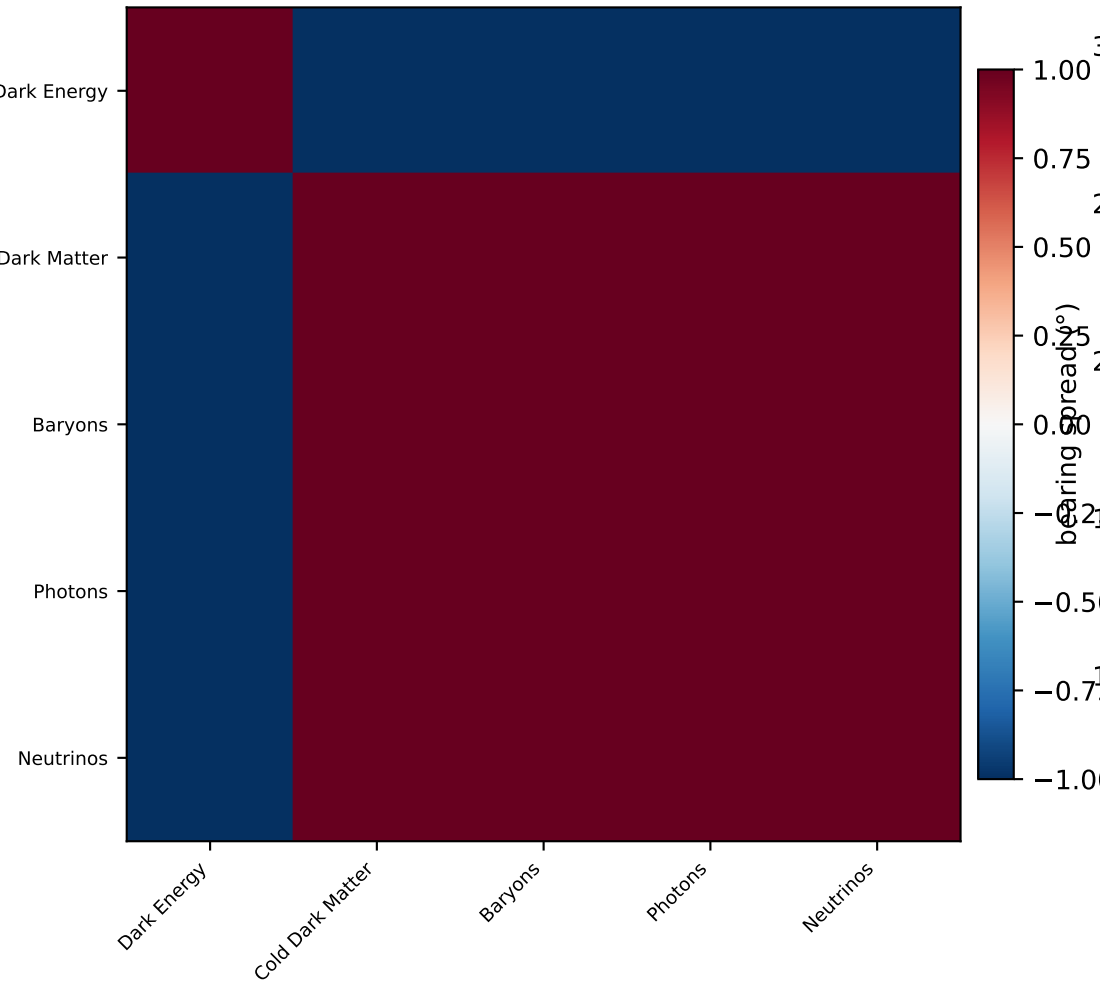
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

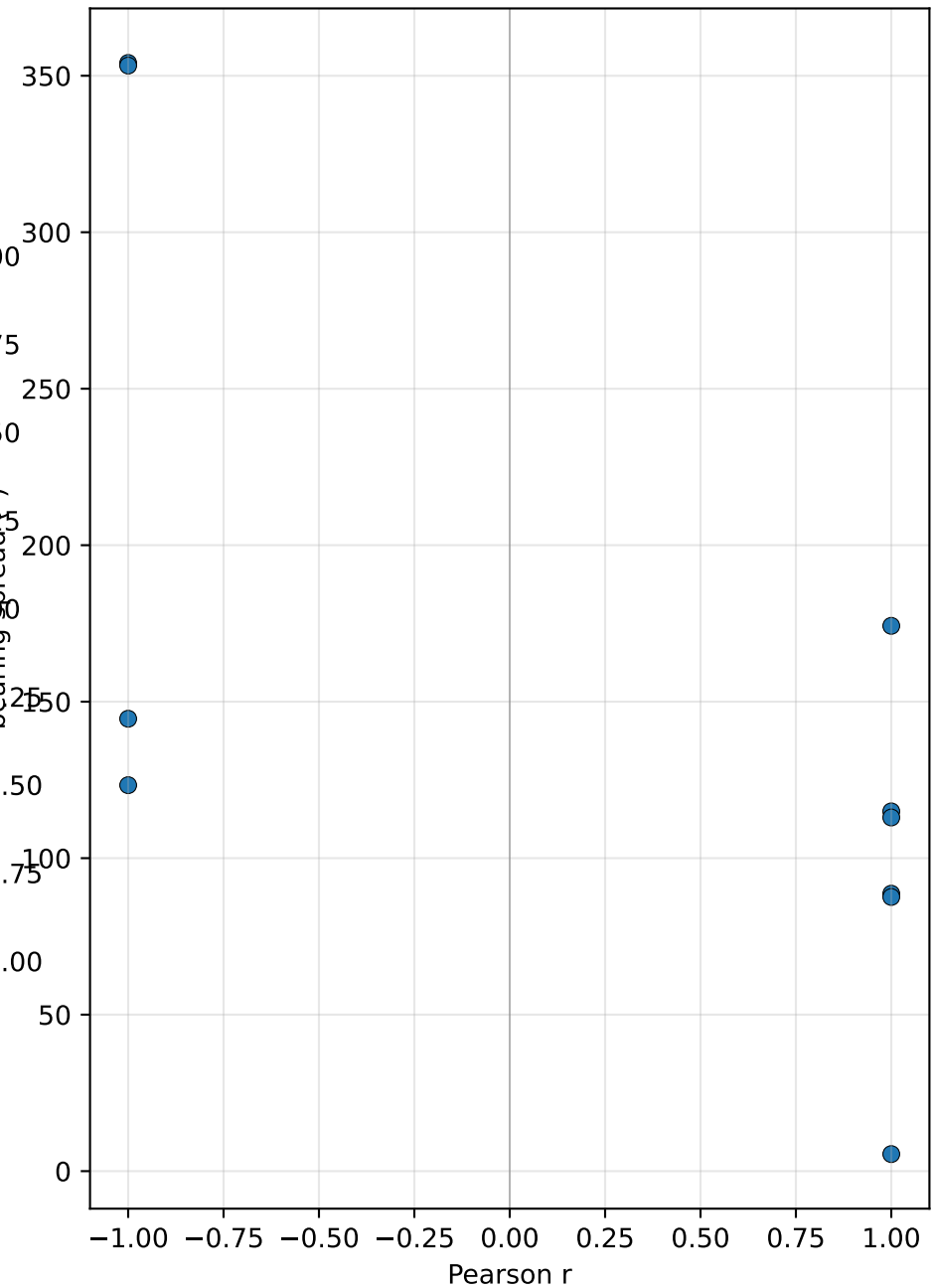


Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

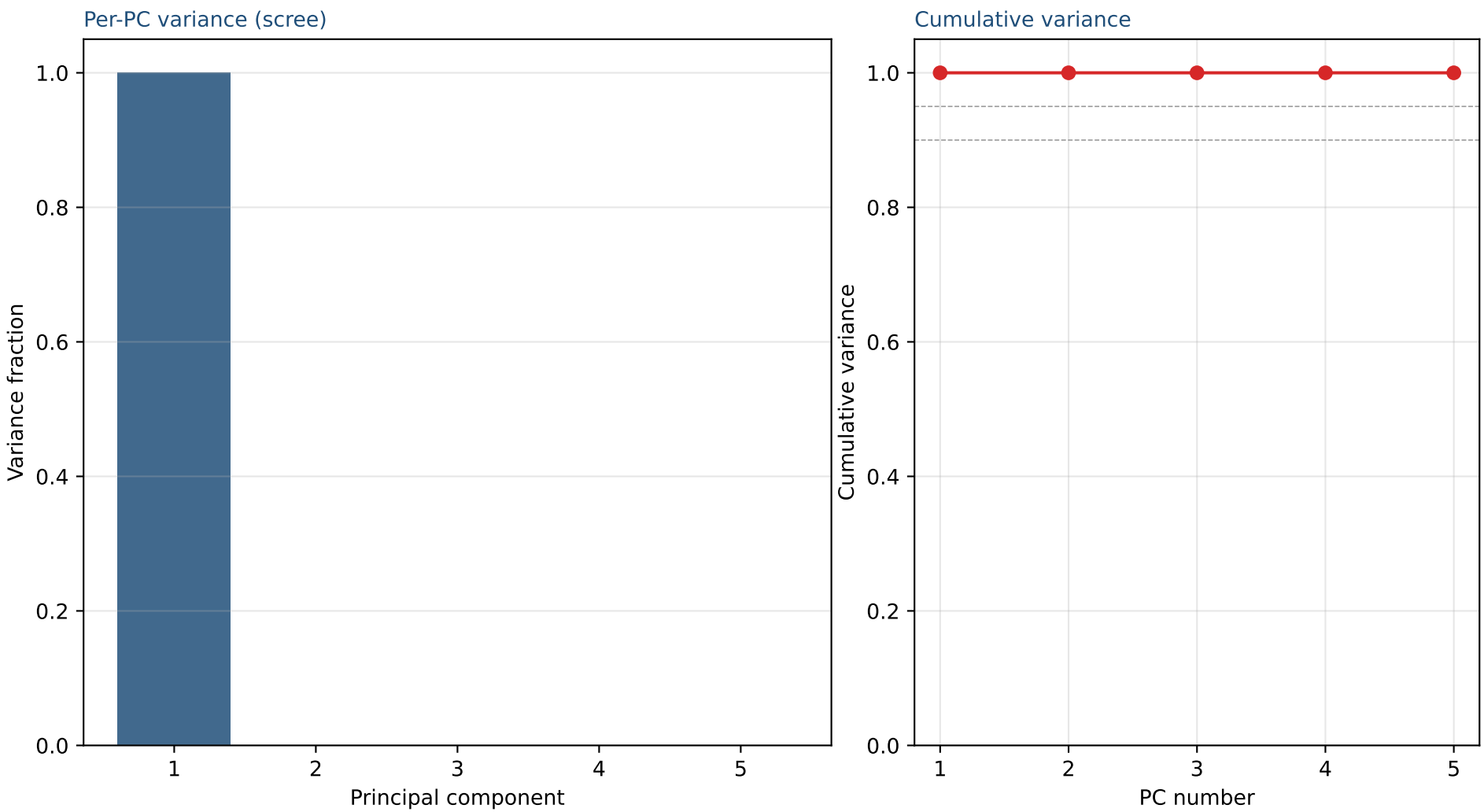




Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 8 | Scree Plot - CLR PCA

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

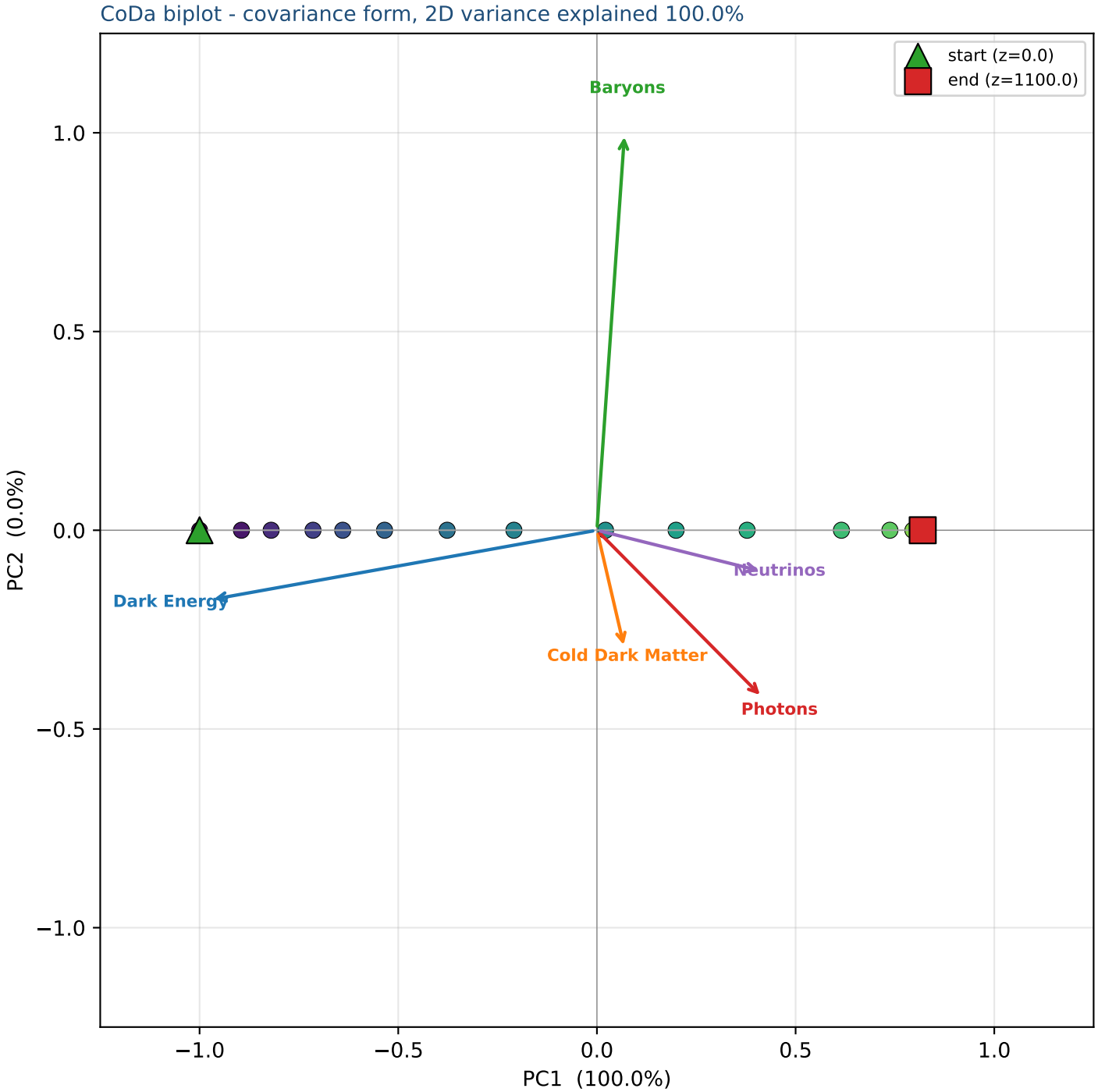



```
Per-PC variance contribution:
  PC1: 100.00%   cum 100.00%
  PC2:  0.00%   cum 100.00%
  PC3:  0.00%   cum 100.00%
  PC4:  0.00%   cum 100.00%
  PC5:  0.00%   cum 100.00%
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 100.0% of variance.
PC2 captures 0.0% of variance.
Total 2D plane:  100.0%
```

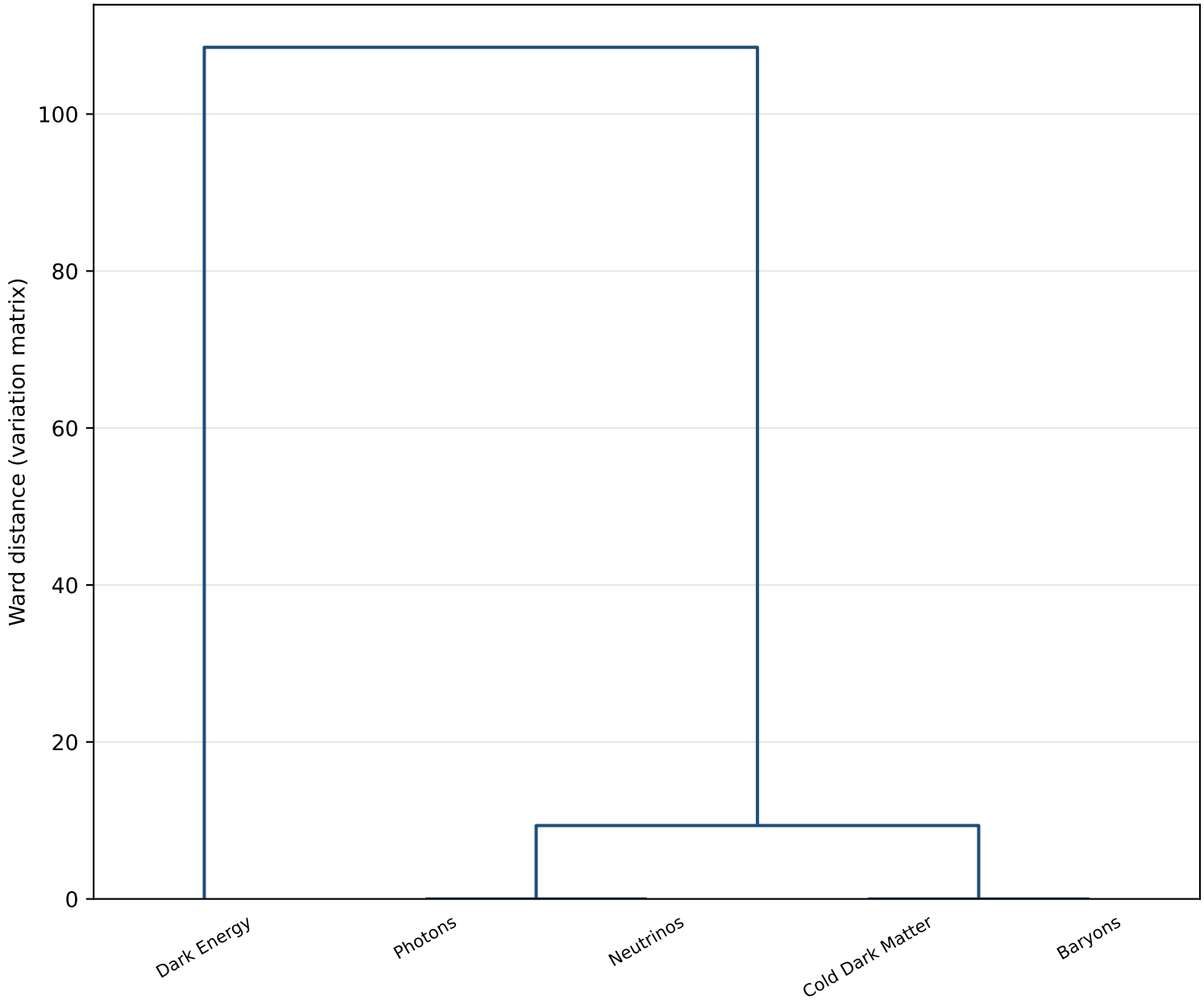
D = 5 ≤ N = 5 — full biplot already shows every carrier (see preceding plate). This page is informational.

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 5 carriers → D-1 = 4 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

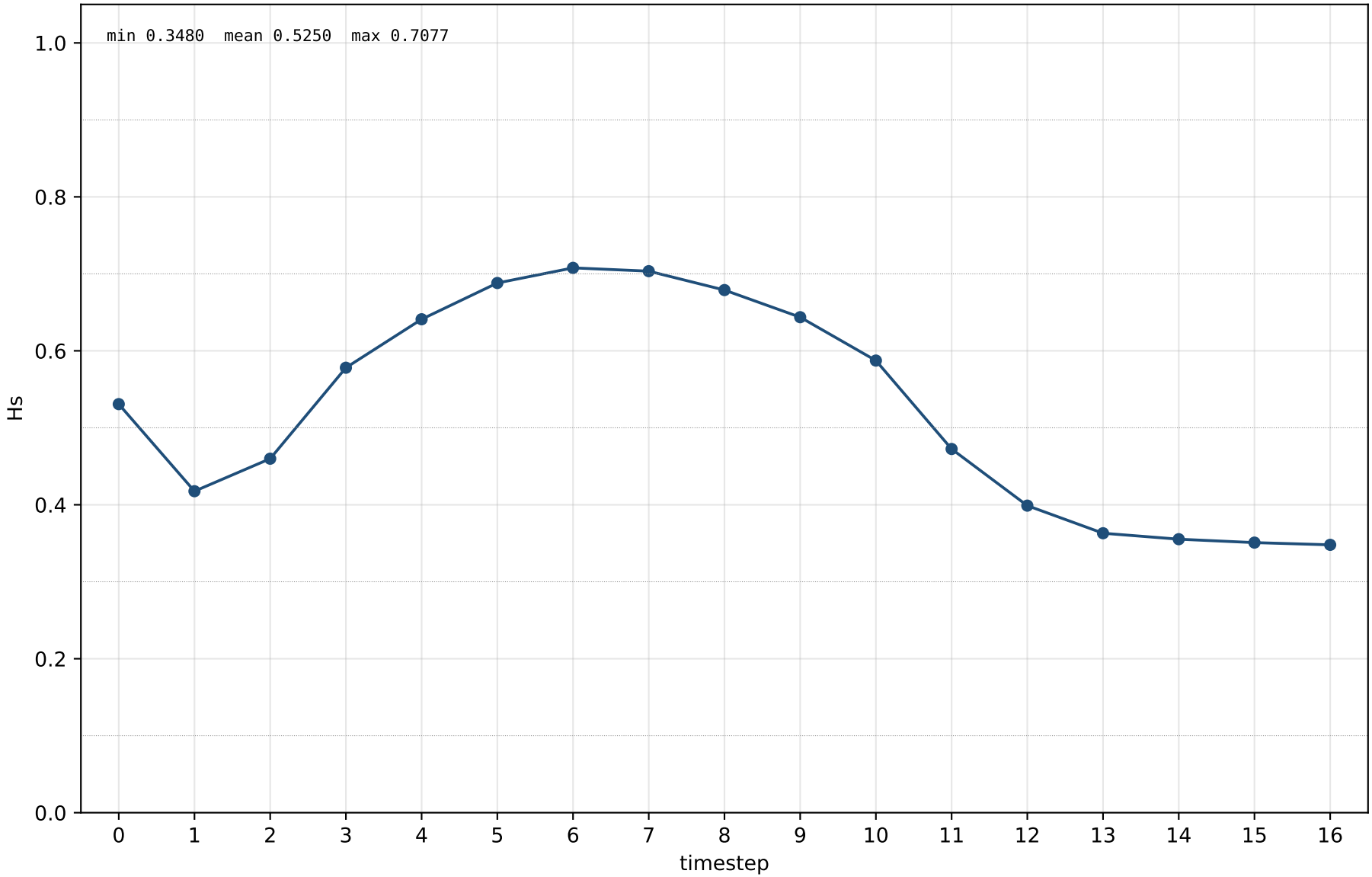
	b1	b2	b3	b4
ergy	.	.	.	+
tter	.	+	-	-
yons	.	-	-	-
tons	+	.	+	-
inos	-	.	+	-

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 5; D-1 = 4 balances.

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

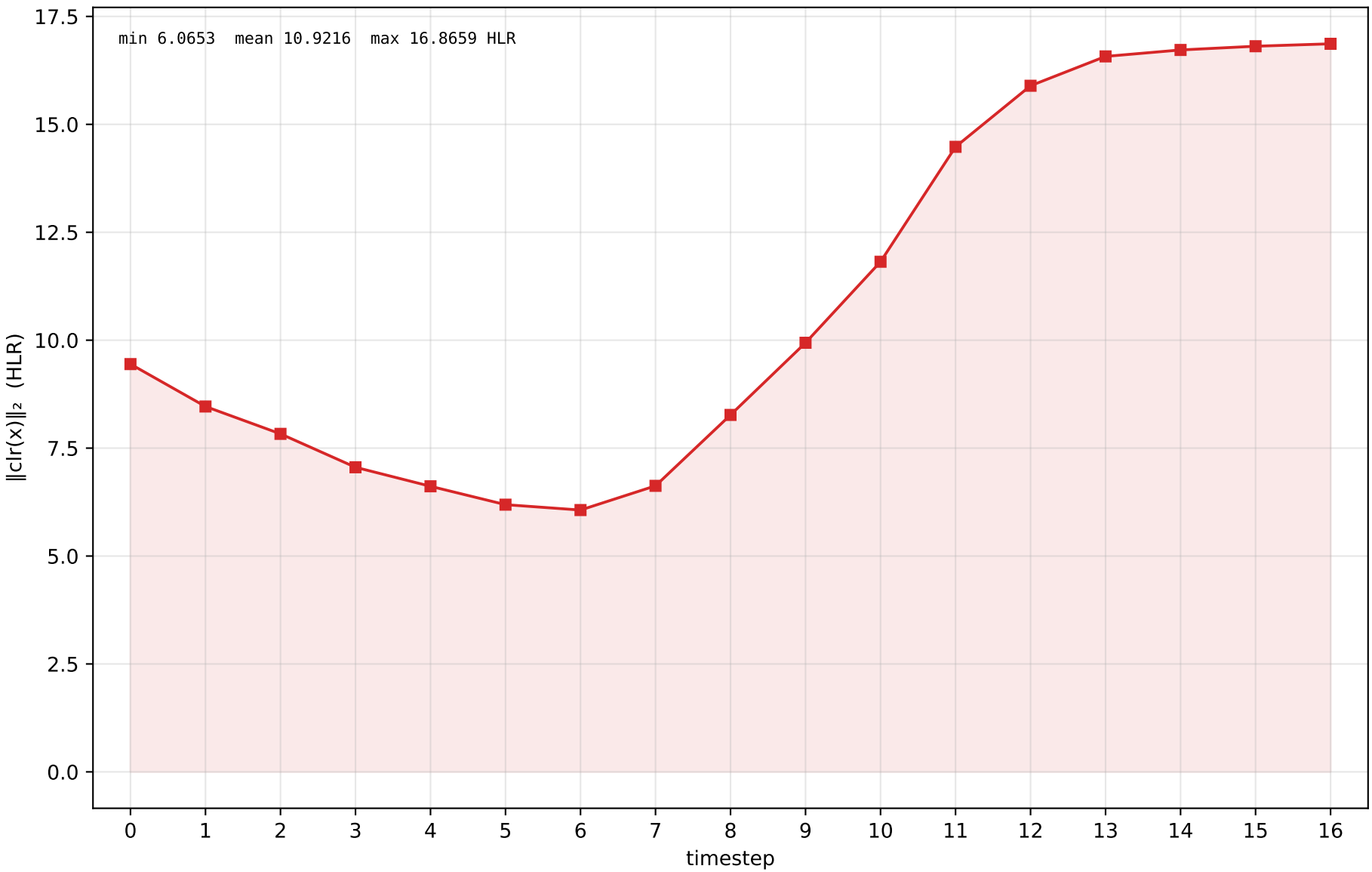
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

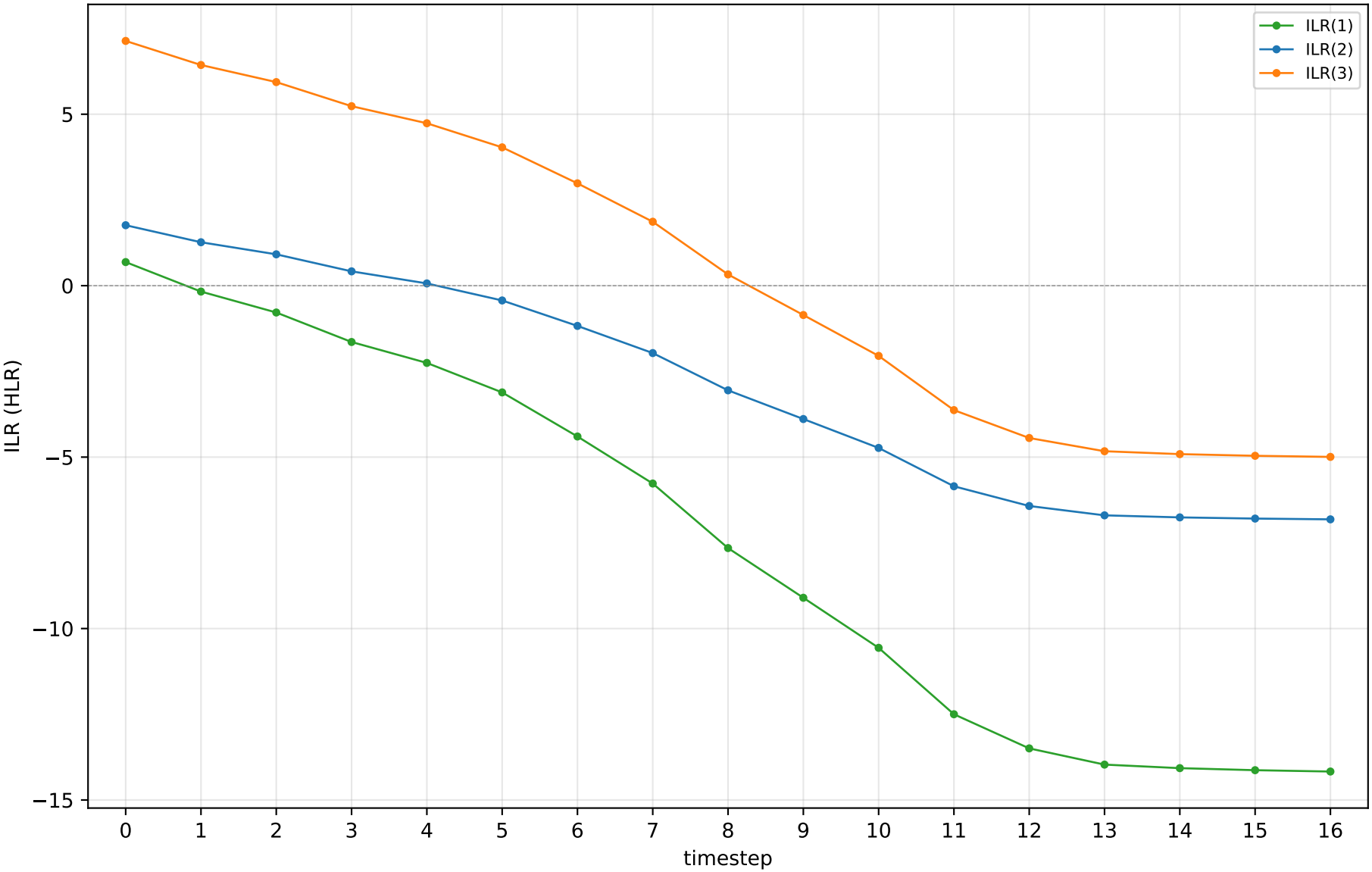
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

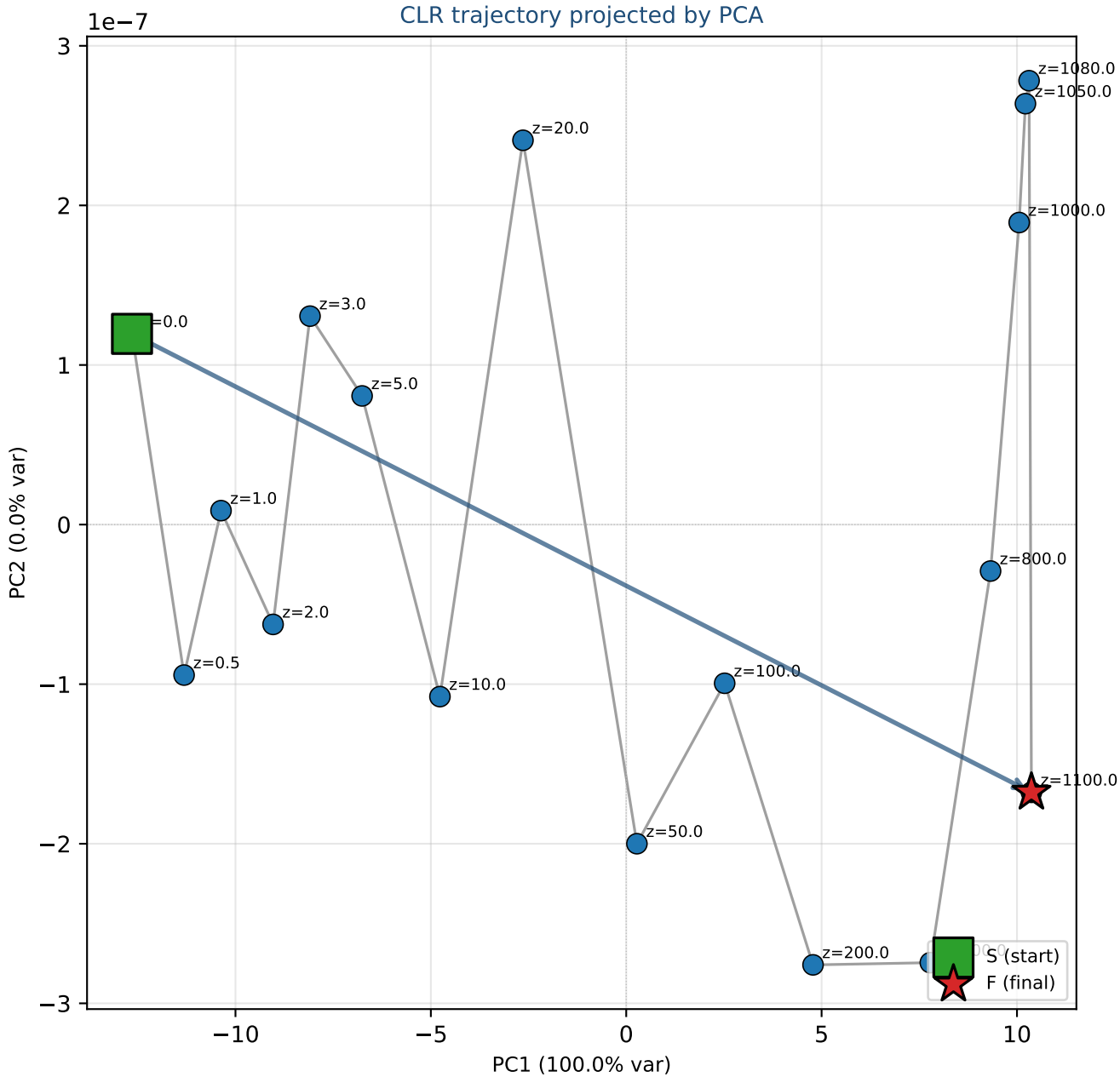
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



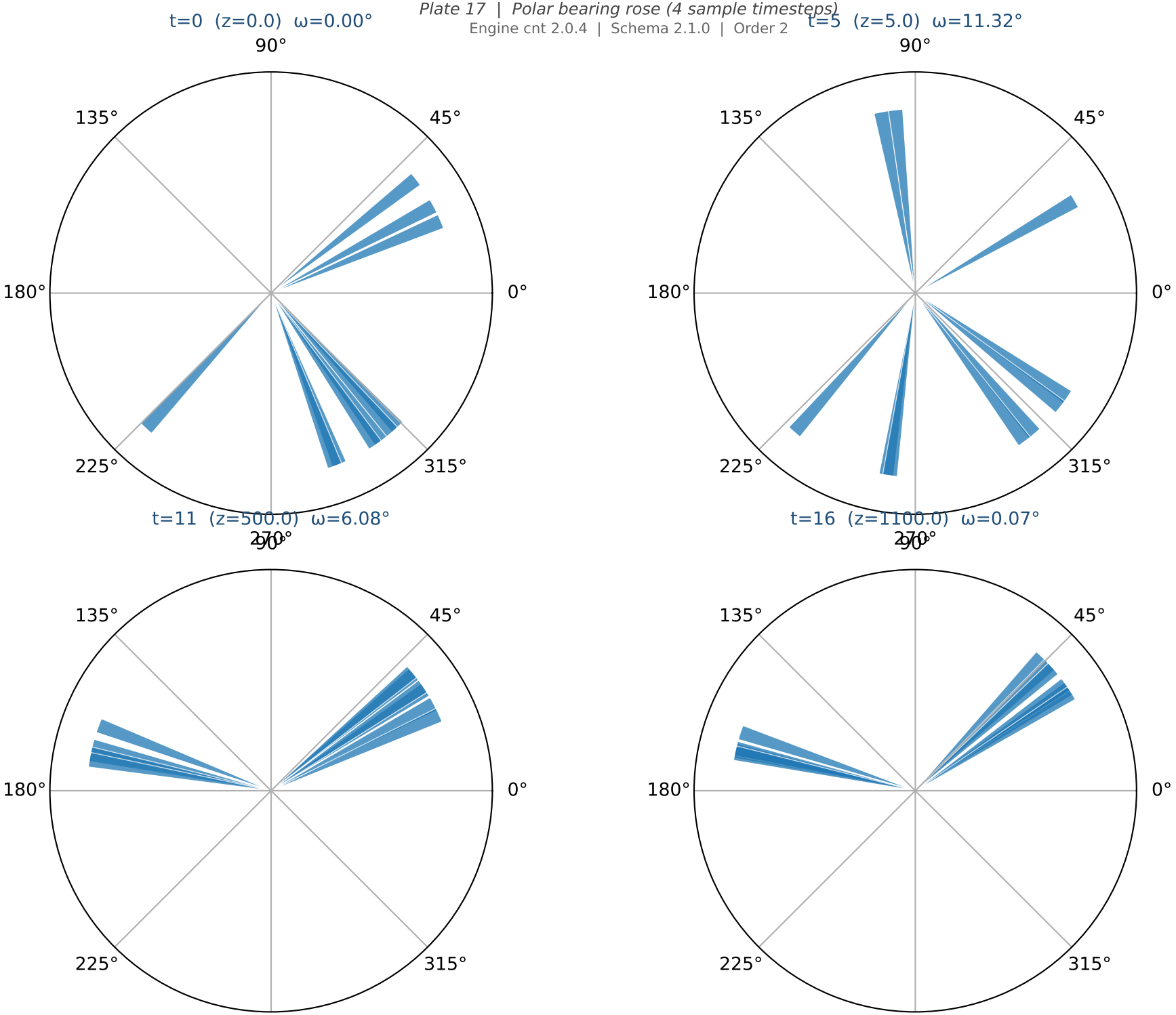
Navigation metrics (V_net = h_F − h_S)

```
h_start label    : z=0.0
h_final label    : z=1100.0

net_distance     : 23.0174 HLR
path_length      : 23.0174 HLR
course_directness: 1.0000
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 9.8532 HLR
dynamic range F  : 20.0394 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 17 | Polar bearing rose (4 sample timesteps)

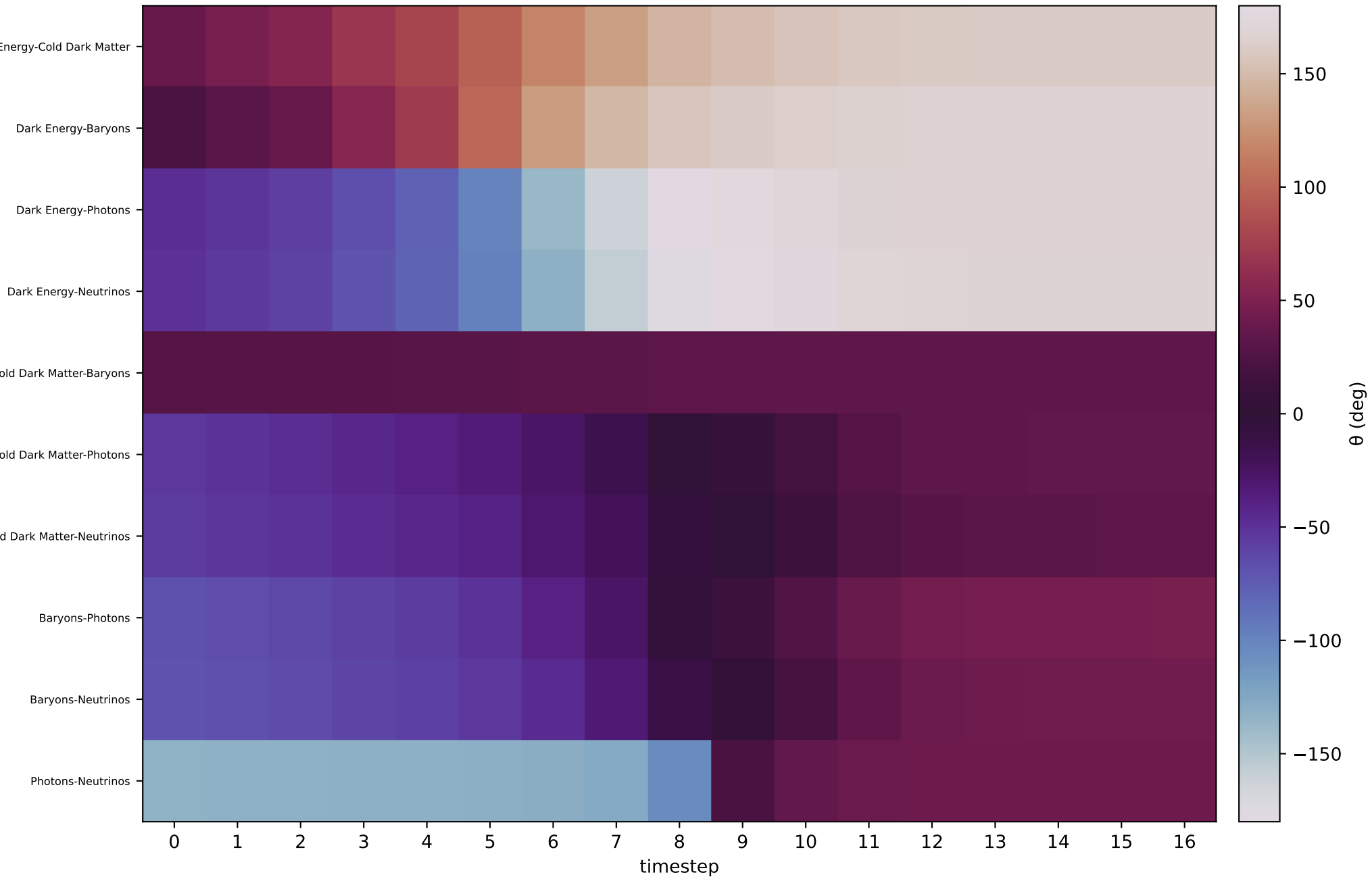
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 18 | Bearing heatmap T × C(D,2) pairs

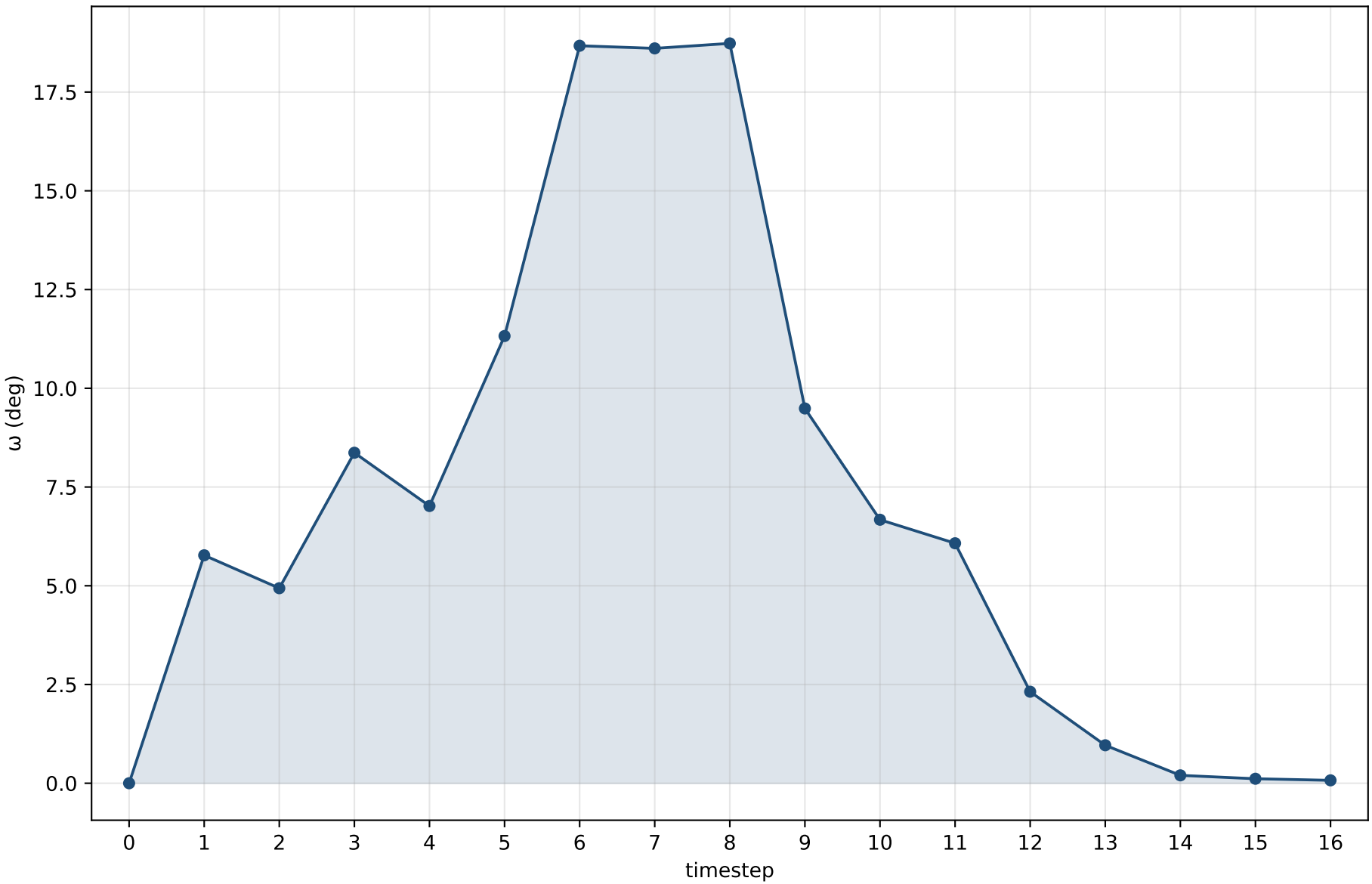
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 19 | ω angular velocity trajectory

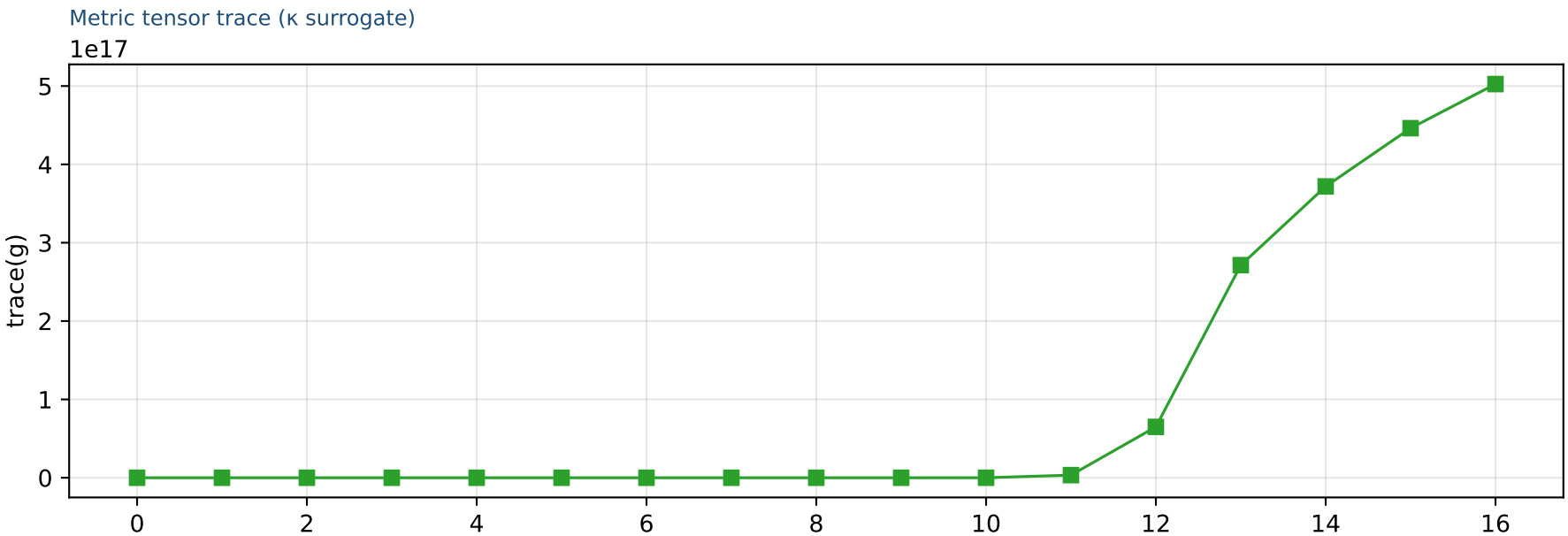
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

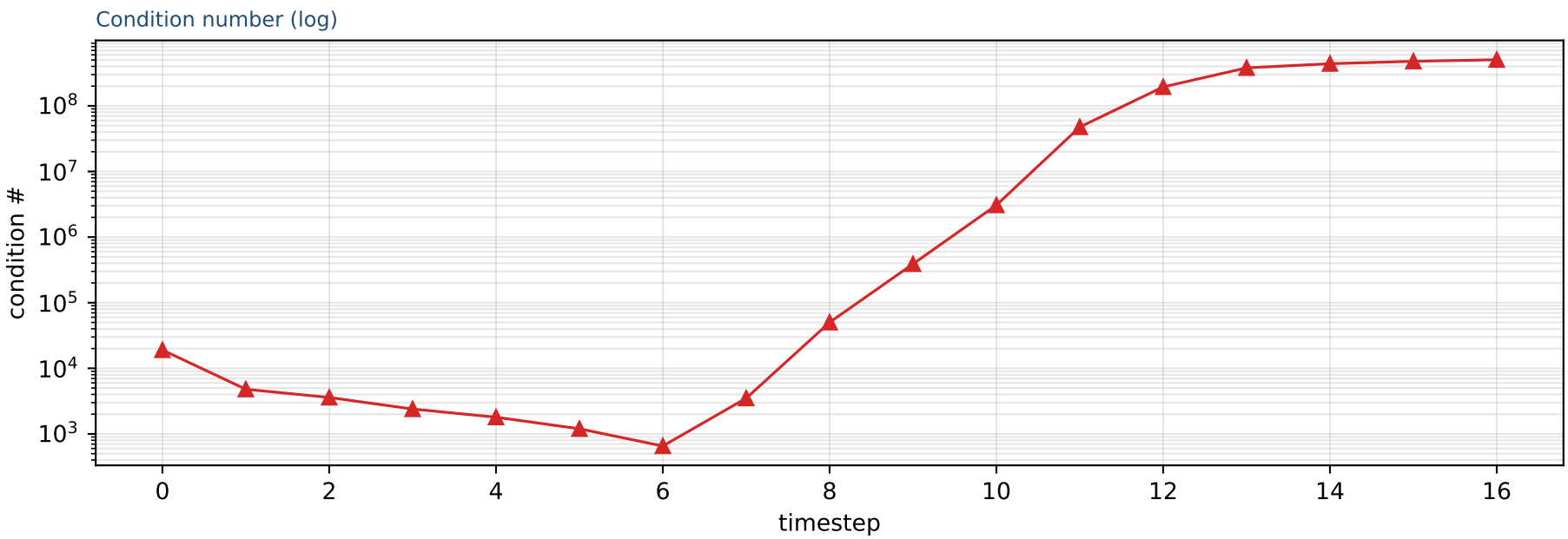


Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 20 | Metric tensor trace + condition number

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

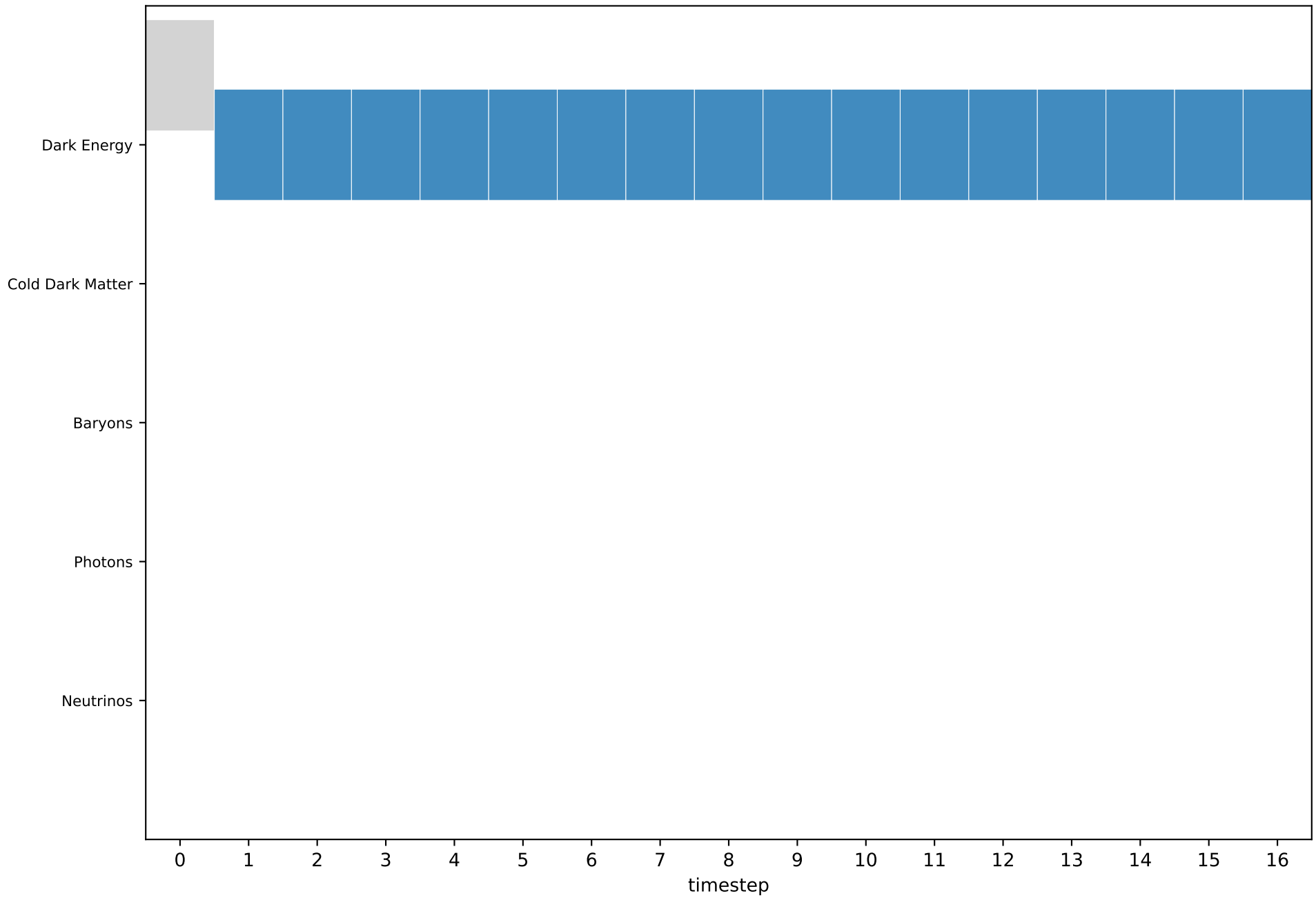




Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 21 | Helmsman σ lineage timeline

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

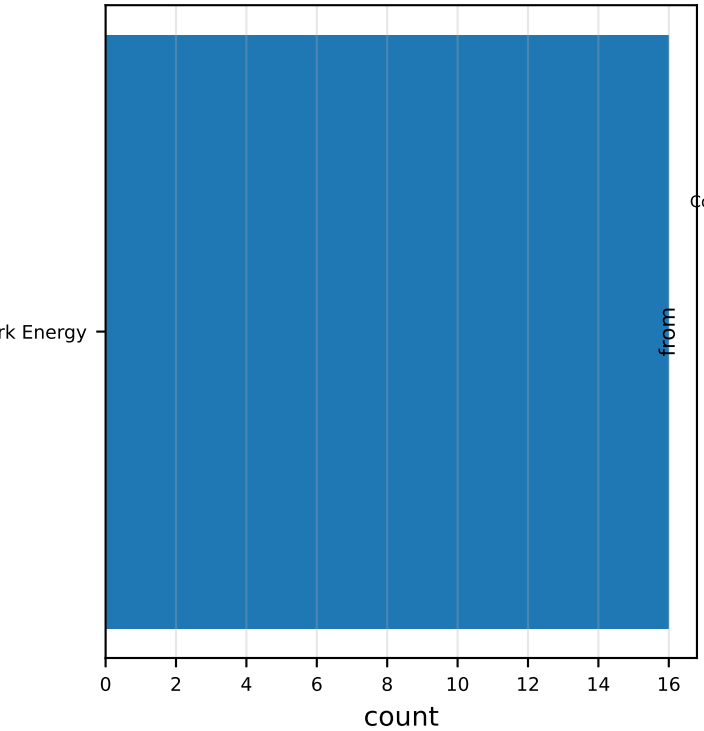


Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

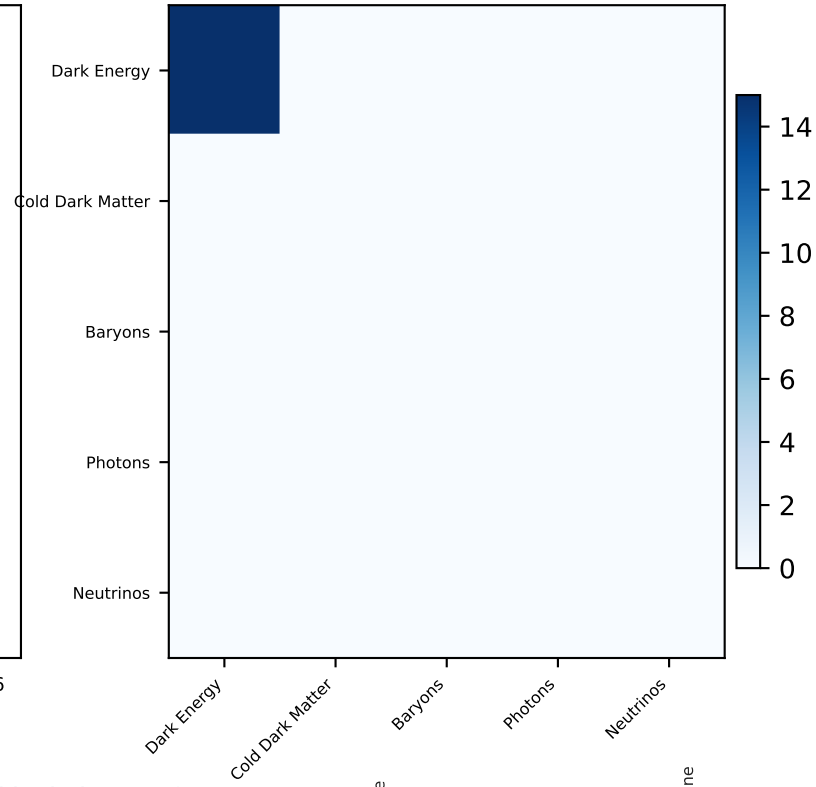
Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

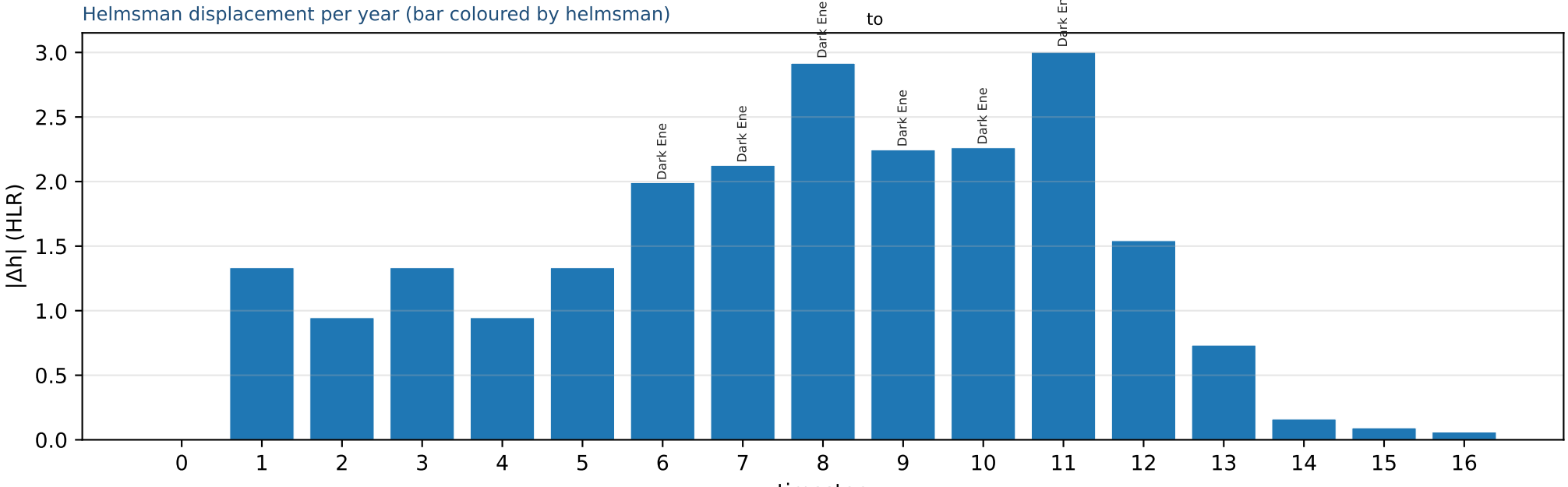
Helmsman frequency



Transition matrix



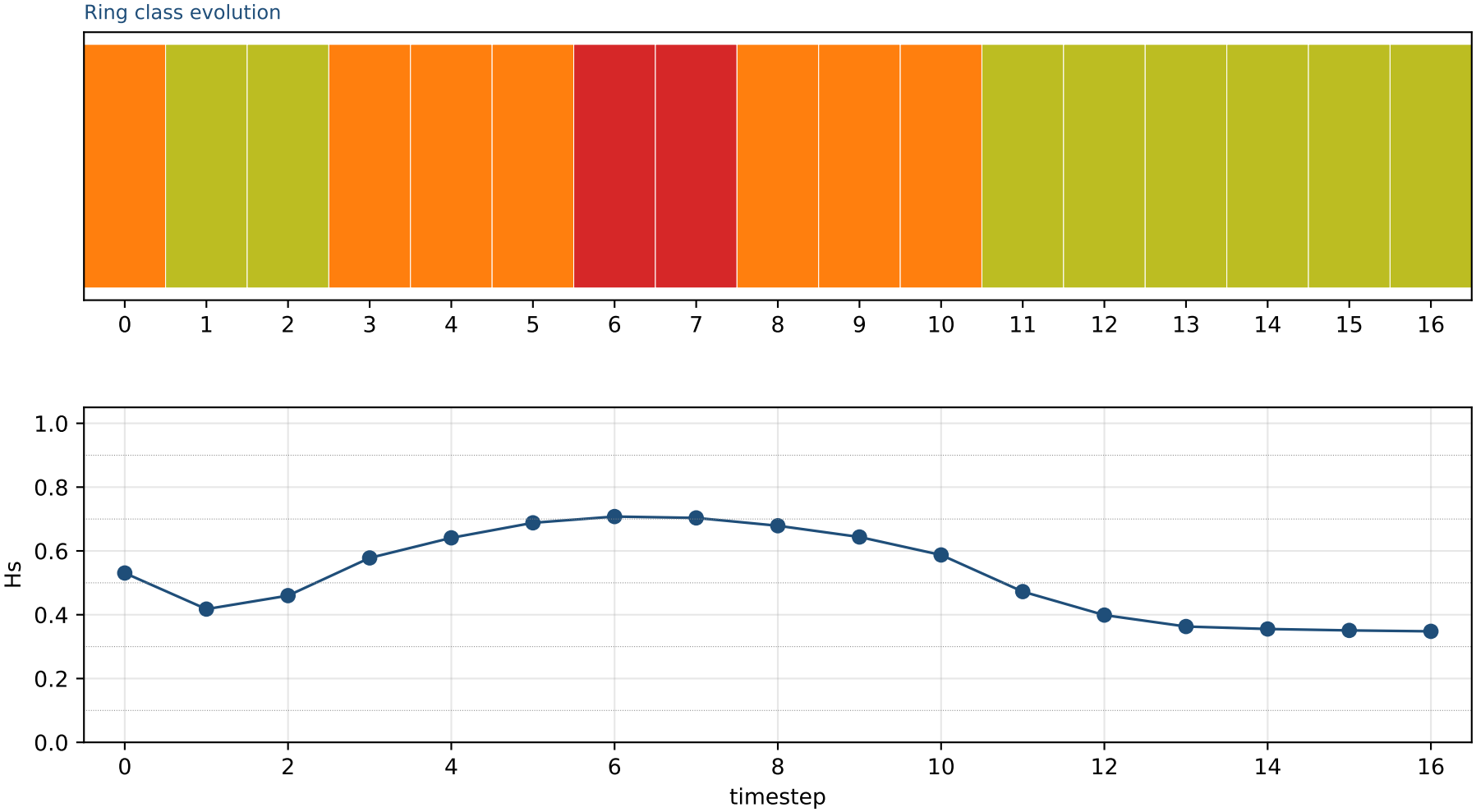
Helmsman displacement per year (bar coloured by helmsman)



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 23 | Ring class evolution + Hs

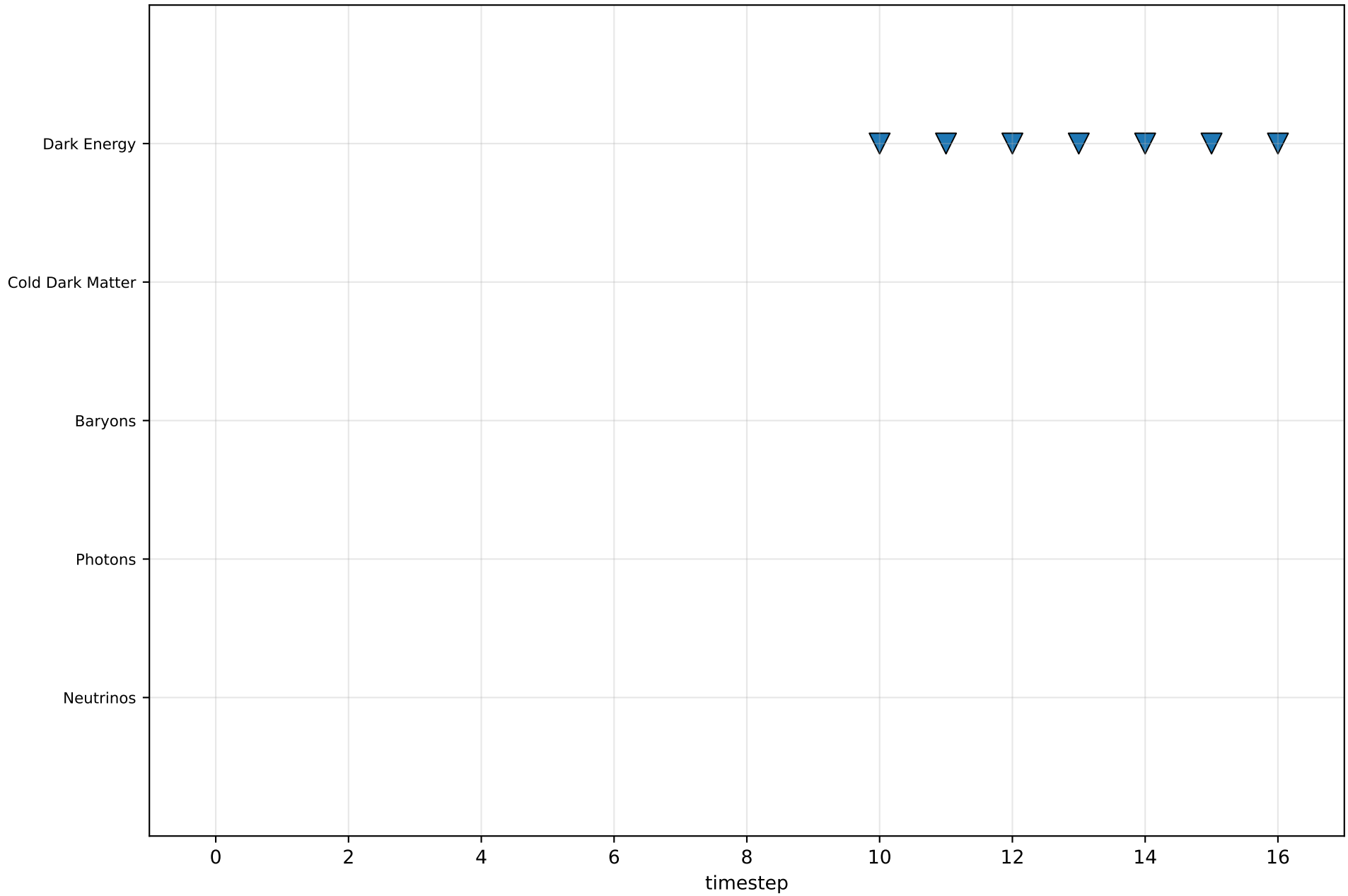
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 24 | Lock event timeline (carriers × timesteps)

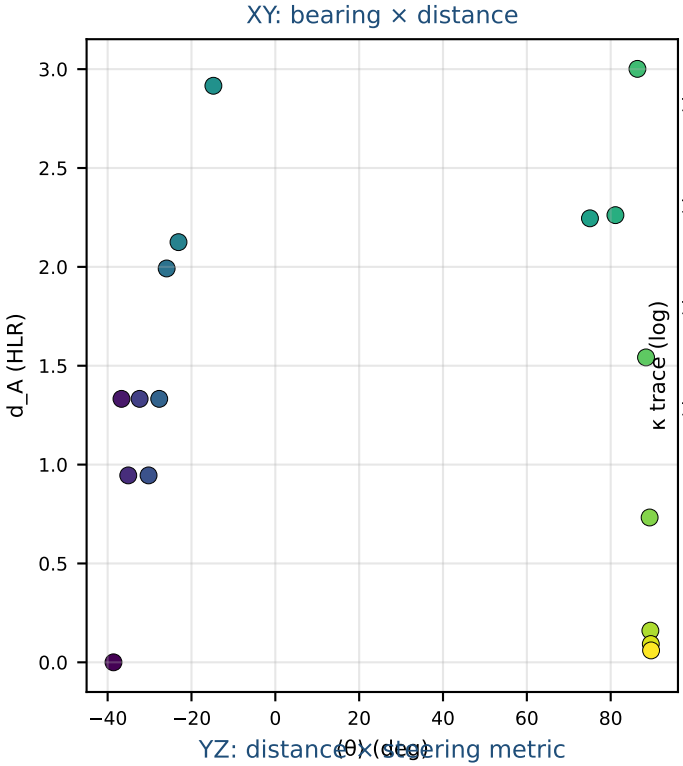
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

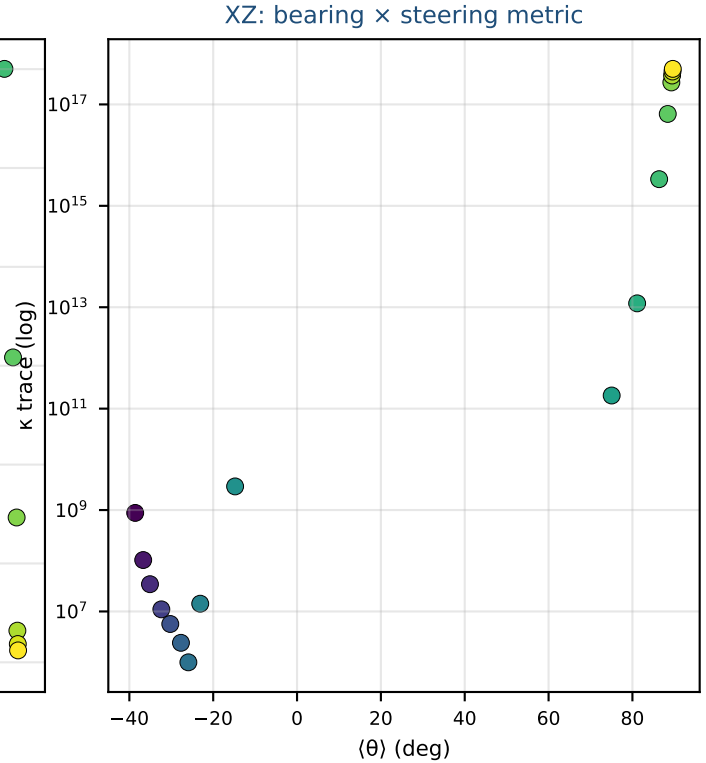


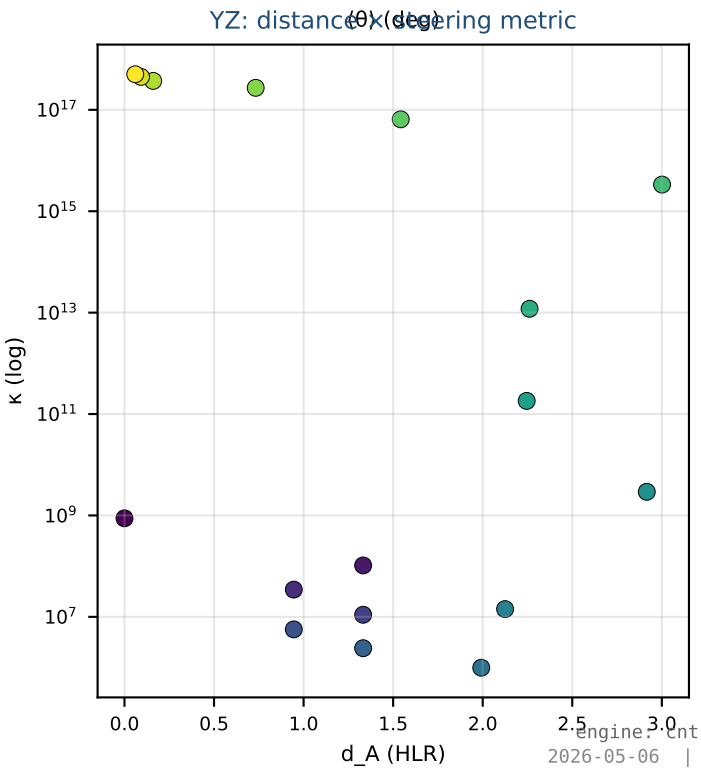
Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2







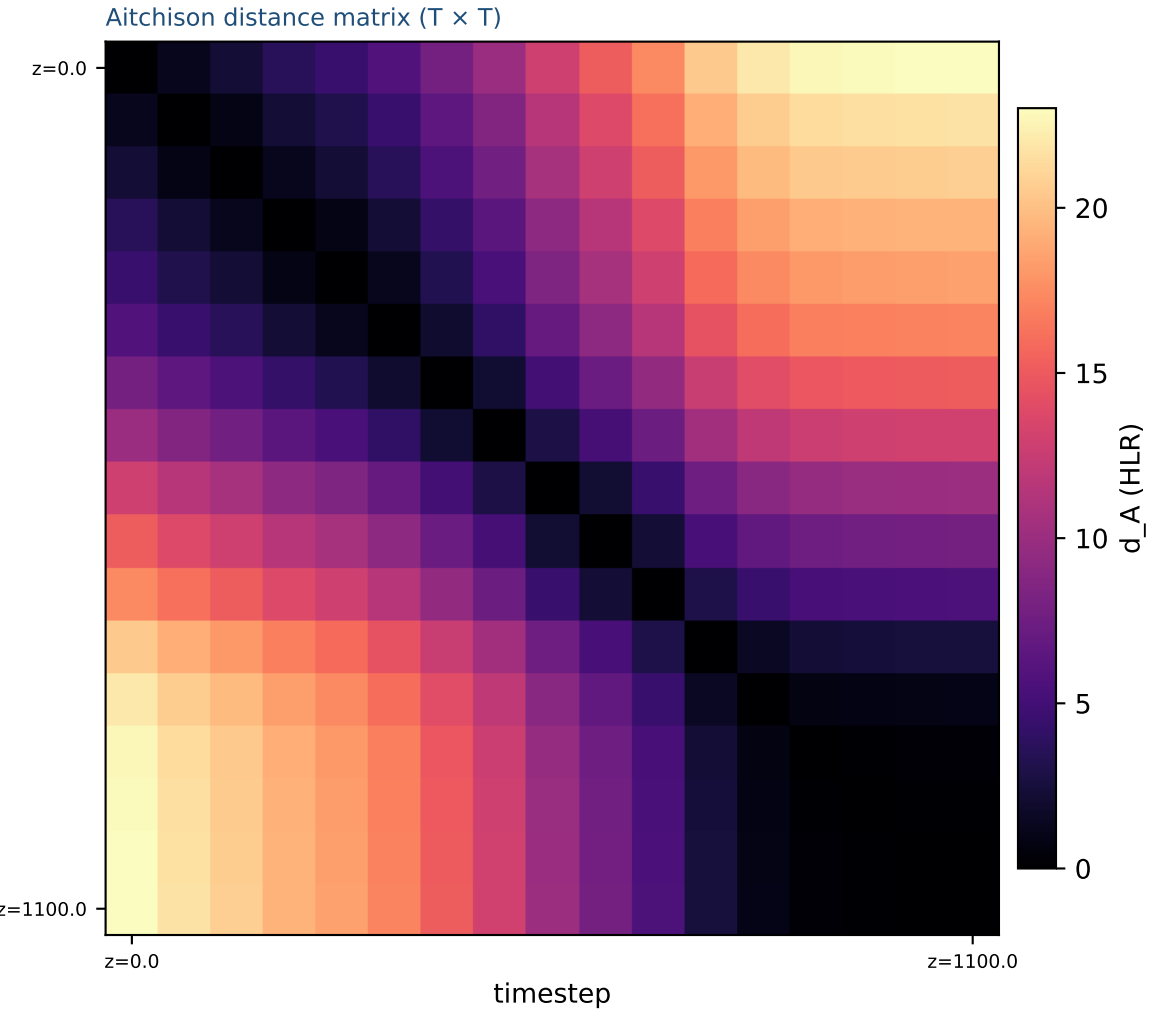
CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [-38.6, 89.6]°
d_A range  : [0.0000, 3.0014] HLR
κ range    : [9.92e+05, 5.02e+17]
Hs range   : [0.3480, 0.7077]
ω max      : 18.73°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

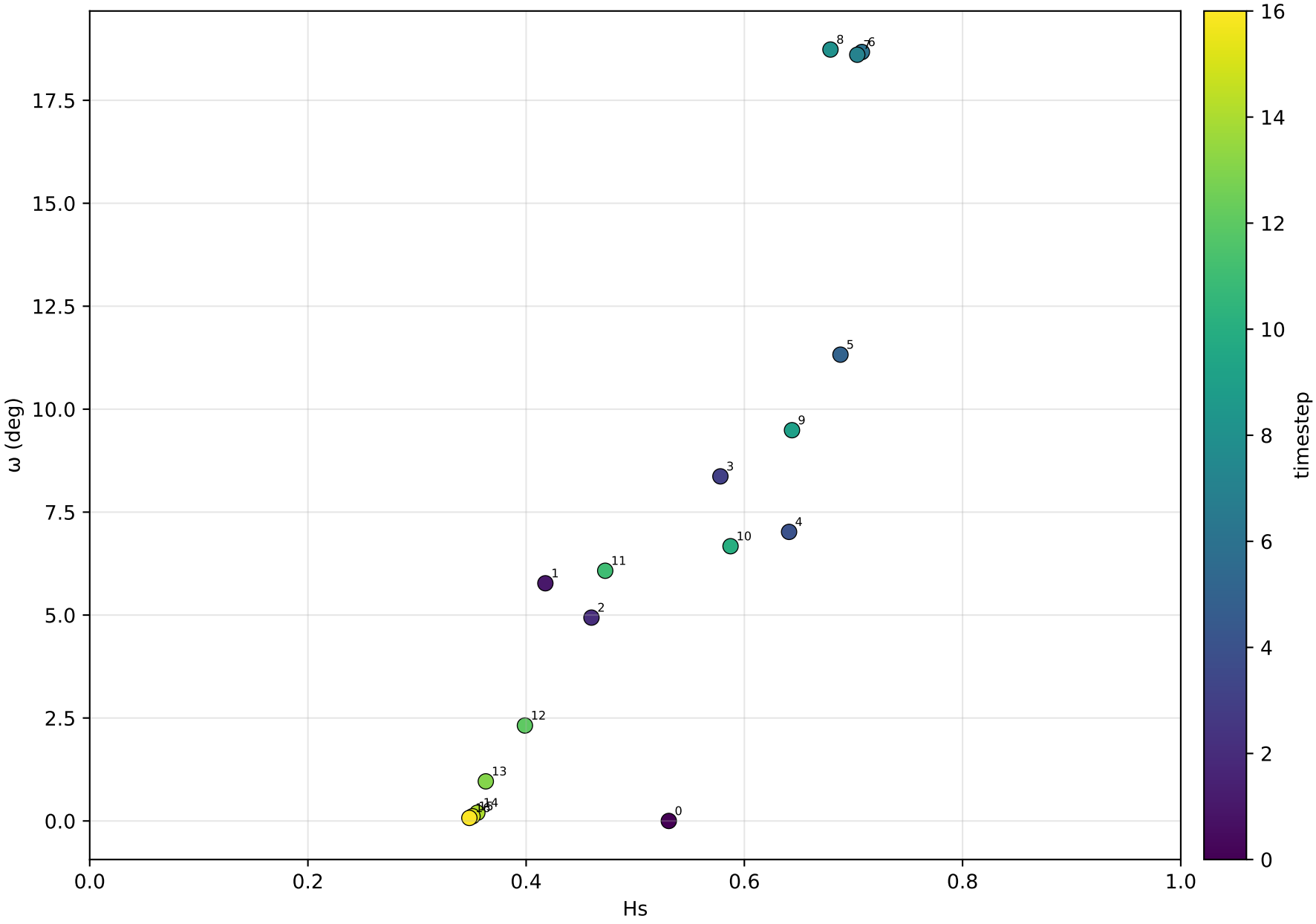


rank	A	B	d_A
1	z=0.0	z=1100.0	23.0174
2	z=0.0	z=1080.0	22.9572
3	z=0.0	z=1050.0	22.8647
4	z=0.0	z=1000.0	22.7045
5	z=0.0	z=800.0	21.9720
6	z=0.5	z=1100.0	21.6849
7	z=0.5	z=1080.0	21.6247
8	z=0.5	z=1050.0	21.5322
9	z=0.5	z=1000.0	21.3720
10	z=1.0	z=1100.0	20.7395
11	z=1.0	z=1080.0	20.6792
12	z=0.5	z=800.0	20.6395
13	z=1.0	z=1050.0	20.5868
14	z=0.0	z=500.0	20.4299
15	z=1.0	z=1000.0	20.4266

Compositional system geometry and dynamics for, ESA Planck cosmic energy budget × redshift z=0.0 → z=1100.0

Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



```
T = 17,  D = 5
carriers: Dark Energy, Cold Dark Matter, Baryons, Photons, Neutrinos

Navigation (System Course Plot):
  net_distance     = 23.0174 HLR
  path_length      = 23.0174 HLR
  course_directness= 1.0000

ω (deg)        : min 0.00  mean 7.02  max 18.73
Hs             : min 0.3480  mean 0.5250  max 0.7077
Aitchison norm : min 6.0653  mean 10.9216  max 16.8659
κ trace        : min 992292.08  max 502489274624892480.00
condition #    : min 6.54e+02  max 5.05e+08

Bearings: range [-178.70, 178.15]°,  std 92.13°
Lock events    : 7
Pairs          : 10
|r| > 0.7 pairs: 10

Helmsman frequency (top 5):
  Dark Energy     16  (94.1%)
  (none)           1  (5.9%)

Variation matrix trace : 0.0000
```